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Termination results for hybrid approach of Joint Spectral Radius computation

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Contents

1. Introduction and Background **6**
1.1. Notation and Preliminaries . . . 7
1.2. Motivation and Applications of the Joint Spectral Radius . . . 7
1.3. Fundamental Theoretical Results . . . 9
1.4. Preprocessing and Normalization . . . 15

2. Invariant Polytope Algorithm **17**
2.1. Extremal and Invariant Norms . . . 17
2.2. Algorithmic Structure . . . 19
2.3. Stopping Criterion and Candidate Rejection . . . 20
2.4. Termination Conditions . . . 20
2.5. Rebalancing and Additional Initial Vertices . . . 20
2.6. Complex Variants . . . 21

3. Finite-Tree Algorithm **22**
3.1. Tree-Based Notation and Definitions . . . 22
3.2. Algorithmic Structure . . . 24
3.3. Termination Conditions . . . 26

4. Hybrid Polytope-Tree Approach **27**
4.1. Hybridization Strategies . . . 27
4.2. Problem Formulation . . . 27
4.3. Hybrid Algorithm . . . 28
4.4. Termination Theory . . . 28
4.4.1. Proof Strategy . . . 28
4.4.2. Auxiliary Termination Results . . . 30
4.4.3. Main Termination Result . . . 31
4.5. Remarks . . . 31
4.5.1. Eventual Polytope Invariance . . . 31
4.6. Implementation Considerations . . . 31

5. Numerical Experiments **33**
5.1. Runtime Analysis of Algorithm 3 on Generated Matrix Families . . . 33
5.2. Comparison with Algorithm 1 . . . 33
5.3. Comparison with Algorithm 2 . . . 33
5.4. Case Studies on Previously Open Problems . . . 33

6. Conclusions and Outlook **34**
6.1. Summary of the Hybrid Framework 34
6.2. Resolved Problem Classes 34
6.3. Computational Efficiency 34
6.4. Remaining Open Problems 34
6.5. Directions for Future Research 34
6.5.1. Extension to the Complex Setting 34
6.5.2. Structured Nonnegative Matrix Families 34
6.5.3. Optimization of Tree Construction and Vertex Selection 34

A. Notation Glossary **35**
A.1. Constants and Globally Defined Variables 35
A.2. Products and Multi-Indices 35
A.3. Norms and JSR 36
A.4. Algorithmic Notation 36
A.5. Tree Terminology 37

Zusammenfassung

Diese Arbeit präsentiert einen hybriden Ansatz zur Berechnung des gemeinsamen Spektralradius (JSR) einer endlichen Menge von Matrizen. Die vorgeschlagene Methode verbindet den invarianten Polytop-Algorithmus und den Endlicher-Baum-Algorithmus, um deren jeweilige Stärken zu nutzen. Die allgemeine Anwendbarkeit des baumbasierten Ansatzes wird mit der Effizienz der polytopbasierten Techniken kombiniert. Der entwickelte Algorithmus konstruiert einen Baumgraphen und überprüft maximale Produkte mithilfe angepasster Normen. Damit kann das Verfahren in Fällen terminieren, in denen rein polytop- oder rein baumbasierte Methoden praktisch scheitern, etwa wegen hoher Laufzeit. Theoretische Resultate zu Korrektheit und Terminierungsbedingungen werden ebenfalls angegeben. Zusätzlich werden JSR-Berechnungen für Matrizenmengen gezeigt, bei denen bestehende Methoden scheitern oder ineffizient sind.

Abstract

This thesis presents a hybrid approach for computing the joint spectral radius (JSR) of a finite set of matrices. The proposed method combines the invariant polytope algorithm with the finite-tree algorithm and uses their complementary strengths. The developed algorithm constructs a structured tree representation and verifies spectrum-maximizing products with adapted norms. This allows termination in cases where standalone polytope-based or tree-based methods are impractical or fail. The thesis also provides theoretical results for correctness and termination conditions.

Keywords: Joint spectral radius, hybrid algorithm, invariant polytope, finite tree method, matrix norms.

1. Introduction and Background

The *Joint Spectral Radius* (JSR) was first introduced by G.-C. Rota and G. Strang in 1960 (Rota and Strang, 1960). They described the JSR as the maximal exponential growth rate of a product of matrices taken from a finite set. Since its inception, the JSR has become a cornerstone in various mathematical and engineering disciplines due to its ability to encapsulate the asymptotic behavior of matrix products.

The concept gained significant traction in the 1990s when researchers began exploring its theoretical properties and practical implications. Notable advancements include its application in wavelet theory, where it assists in the construction of refinable functions (Daubechies and Lagarias, 1992), as well as in control theory, where it is used to analyze the stability of switched linear systems (Jungers, 2009), for which we provide a small example below. The computational challenges associated with determining the JSR have inspired the development of several algorithms, such as the invariant-polytope method (Guglielmi and Protasov, 2013) and the finite-tree method (Möller and Reif, 2014).

Despite the progress, JSR computation remains challenging, particularly due to the exponential complexity of exploring all possible matrix products. This thesis contributes to this line of work by combining the invariant-polytope algorithm and the finite-tree algorithm into a hybrid approach that mitigates their respective limitations.

Thesis Outline

This thesis is structured as follows: Chapter 1 provides background on the JSR and its basic properties. Chapters 2 and 3 present the key ideas behind the algorithms used to construct the proposed hybrid approach, including their theoretical foundations and algorithmic implementations. Chapter 4 discusses possible combinations of the former approaches, proposes the so-called Tree-flavored invariant-polytope algorithm, and provides proofs of its termination properties, which form the main results of this thesis. Chapter 5 presents numerical results to analyze the efficiency and applicability of the hybrid algorithm. Chapter 6 concludes with insights and future directions. Appendix A collects a complete notation glossary for quick reference.

1.1. Notation and Preliminaries

Unless stated otherwise, all matrices are real $d \times d$ matrices and all matrix norms are submultiplicative. We use the following notation conventions consistently across all chapters:

- Matrix families and dimensions:

$$\mathcal{A} = \{A_1, \ldots, A_n\}, \qquad \tilde{\mathcal{A}} = \{\tilde{A}_1, \ldots, \tilde{A}_n\}, \qquad d = \text{matrix dimension}, \quad n = |\mathcal{A}|.$$

- Generator notation:

$$\mathcal{G} = \{G_1, \ldots, G_m\}, \qquad m = |\mathcal{G}|.$$

- Positive indices and products:

$$\mathcal{I} := \{1, \ldots, n\}, \quad I = [i_1, \ldots, i_k] \in \mathcal{I}^k, \quad A_I := A_{i_k} \cdots A_{i_1}, \quad \mathcal{A}^k := \{A_I : I \in \mathcal{I}^k\}.$$

- Scalars and spectra: α denotes a scalar, x a vector, λ an eigenvalue, $\rho(\cdot)$ the spectral radius, and $\sigma(\cdot)$ a singular value.
- Polytope norms:

$$\|\cdot\|_{\mathrm{co_s}(V)} = \|\cdot\|_V.$$

Appendix A contains the full notation glossary used throughout the thesis.

1.2. Motivation and Applications of the Joint Spectral Radius

Let $\mathcal{A} = \{A_1, \cdots, A_n\}$ be a finite set of $d \times d$ matrices. Consider a discrete-time linear system generated by a matrix sequence from $\mathcal{A}$:

$$x_{t+1} = B_t x_t, \quad x_0 \in \mathbb{R}^d, \tag{1.1}$$

where $B_t \in \mathcal{A}$ for all $t \in \mathbb{N}$.

A key question is whether the system is *uniformly asymptotically stable*, i.e., whether every trajectory (x_t) converges to zero for all initial states and all matrix sequences $(B_t)_{t\geq 0}$ with $B_t \in \mathcal{A}$:

$$\lim_{t\to\infty} x_t = 0.$$

This stability question is answered by a generalization of the spectral radius for a single matrix, the so-called *joint spectral radius (JSR)* of a matrix set $\mathcal{A}$ (Rota and Strang, 1960).

Definition 1.2.1. (Rota and Strang, 1960) For a finite set of matrices $\mathcal{A} = \{A_1, \ldots, A_n\}$, the JSR is defined as:

$$\mathrm{JSR}(\mathcal{A}) := \lim_{k\to\infty} \max_{P\in\mathcal{A}^k} \|P\|^{1/k}. \tag{1.2}$$

where $\|\cdot\|$ denotes any matrix norm.

This definition quantifies the maximal asymptotic growth rate of matrix products from $\mathcal{A}$.

Illustrative Example

Consider the matrix set:

$$\mathcal{A} = \left\{ A_1 = \begin{bmatrix} 0.5 & 1 \\ 0 & 0.5 \end{bmatrix}, \quad A_2 = \begin{bmatrix} 0.3 & 0 \\ 1 & 0.3 \end{bmatrix} \right\}.$$

Each matrix individually has spectral radius less than 1. However, their product

$$A_1 A_2 = \begin{bmatrix} 0.5 & 1 \\ 0 & 0.5 \end{bmatrix} \begin{bmatrix} 0.3 & 0 \\ 1 & 0.3 \end{bmatrix} = \begin{bmatrix} 1.15 & 0.3 \\ 0.5 & 0.15 \end{bmatrix}$$

has spectral radius of approximately

$$\rho(A_1 A_2) \approx 1.2825 > 1.$$

This example shows that switching can amplify the effects of individual matrices, leading to divergent behavior even if all single matrices are contractive. Thus, the JSR can exceed the spectral radius of any individual matrix in the set.

The following results motivate the importance of bounding or approximating $\mathrm{JSR}(\mathcal{A})$ (Jungers, 2009):

- The system (1.1) is *uniformly bounded* (i.e., $\sup_t \|x_t\| < \infty$ for all $x_0 \in \mathbb{R}^d$) if and only if $\mathrm{JSR}(\mathcal{A}) \leq 1$ and $\mathcal{A}$ is irreducible.
- It is *uniformly asymptotically stable* if and only if $\mathrm{JSR}(\mathcal{A}) < 1$.

Understanding the joint spectral radius is therefore critical for analyzing the asymptotic behavior of switched linear systems and motivates reliable computational methods (Jungers, 2009). Beyond control theory, the JSR also appears in approximation theory, for example in regularity questions for two-scale and refinement equations (Daubechies and Lagarias, 1992; Maesumi, 2000). Related spectral-growth ideas are used in coding-theoretic capacity problems (Blondel et al., 2006) and in discrete mathematics and combinatorics (Protasov, 2000; Jungers et al., 2008, 2009). These links motivate efficient computational methods, including the hybrid algorithm studied in this thesis (Mejstrik and Reif, 2025).

1.3. Fundamental Theoretical Results

Lemma 1.3.1 (Fekete's Lemma)**.** *Let $(a_n)_{n\geq 1}$ be a sequence of real, nonnegative numbers such that*

$$a_{m+n} \leq a_m + a_n \qquad \text{for all } m, n \geq 1. \tag{1.3}$$

Then the limit

$$\lim_{n\to\infty} \frac{a_n}{n}$$

exists and is equal to $\inf_{n\geq 1}\left(\frac{a_n}{n}\right)$.

Proof. Set $b_n := a_n/n$ and let $\alpha := \inf_{n\geq 1} b_n$. Fix $n \geq 1$ and write any $k \geq 1$ as $k = qn+r$ with integers $q \geq 0$ and $0 \leq r < n$. By subadditivity (1.3),

$$a_k = a_{qn+r} \leq qa_n + a_r.$$

Dividing by k gives

$$\frac{a_k}{k} \leq \frac{qa_n}{qn+r} + \frac{a_r}{qn+r} \leq \frac{a_n}{n} + \frac{a_r}{qn}.$$

Taking $\limsup_{k\to\infty}$ (equivalently $q \to \infty$) yields

$$\limsup_{k\to\infty} \frac{a_k}{k} \leq \frac{a_n}{n}.$$

Since n was arbitrary, we obtain

$$\limsup_{k\to\infty} \frac{a_k}{k} \leq \inf_{n\geq 1} \frac{a_n}{n} = \alpha.$$

On the other hand, by definition of infimum we always have $\alpha \leq a_k/k$ for all k, hence $\alpha \leq \liminf_{k\to\infty} a_k/k$. Therefore $\liminf = \limsup = \alpha$, so the limit exists and equals α. □

Theorem 1.3.2. *(Rota and Strang, 1960; Jungers, 2009) The JSR is well-defined and independent of the choice of the matrix norm.*

Proof. Let $\|\cdot\|_1$ and $\|\cdot\|_2$ be two matrix norms on $\mathbb{R}^{n\times n}$. By equivalence of norms in finite-dimensional vector spaces, there exist constants $c, C > 0$ such that:

$$c\|P\|_1 \leq \|P\|_2 \leq C\|P\|_1 \quad \text{for all } P \in \mathbb{R}^{d\times d}.$$

Therefore, we have

$$\begin{aligned}
&\lim_{k\to\infty} \max_{P\in\mathcal{A}^k} \|P\|_1^{1/k} \leq \lim_{k\to\infty} \left(\frac{1}{c}\right)^{\frac{1}{k}} \max_{P\in\mathcal{A}^k} \|P\|_2^{1/k} \\
&= \lim_{k\to\infty} \max_{P\in\mathcal{A}^k} \|P\|_2^{1/k} \leq \lim_{k\to\infty} C^{\frac{1}{k}} \max_{P\in\mathcal{A}^k} \|P\|_1^{1/k} \\
&= \lim_{k\to\infty} \max_{P\in\mathcal{A}^k} \|P\|_1^{1/k},
\end{aligned}$$

which makes the limit independent of the choice of the used matrix norm. The existence of this limit is a direct consequence of Fekete's Lemma. □

To facilitate the analysis of the algorithms and presentation of the main results, we first establish fundamental properties of the JSR.

Homogeneity of the Joint Spectral Radius

Proposition 1.3.3. *(Jungers, 2009) The JSR is homogeneous, meaning for any set of matrices $\mathcal{A}$ and scalar $\alpha \in \mathbb{R}$ we have*

$$\mathrm{JSR}(\alpha\mathcal{A}) = |\alpha| \,\mathrm{JSR}(\mathcal{A}). \tag{1.4}$$

Proof. Let $\mathcal{A} = \{A_1, \ldots, A_n\}$ and $\alpha \in \mathbb{R}$. Then:

$$\begin{aligned}
\mathrm{JSR}(\alpha\mathcal{A}) &= \lim_{k\to\infty} \max_{A_i\in\mathcal{A}^k} \|\alpha A_{i_k} \cdots \alpha A_{i_1}\|^{1/k} \\
&= |\alpha| \lim_{k\to\infty} \max_{A_i\in\mathcal{A}^k} \|A_{i_k} \cdots A_{i_1}\|^{1/k} \\
&= |\alpha| \,\mathrm{JSR}(\mathcal{A})
\end{aligned}$$

□

Berger–Wang Characterization and Three-Member Inequality

Proposition 1.3.4. *(Berger and Wang, 1992) The* JSR *can equivalently be defined as*

$$\mathrm{JSR}(\mathcal{A}) = \limsup_{k\to\infty} \max_{P\in\mathcal{A}^k} \rho(P)^{1/k},$$

if $\mathcal{A}$ is a finite (or compact) set.

Proposition 1.3.5. *(Rota and Strang, 1960; Jungers, 2009) Let $\|\cdot\|$ be a matrix norm. Then for every $k \in \mathbb{N}$,*

$$\max_{P\in\mathcal{A}^k} \rho(P)^{1/k} \;\leq\; \mathrm{JSR}(\mathcal{A}) \;\leq\; \max_{P\in\mathcal{A}^k} \|P\|^{1/k}. \tag{1.5}$$

Proof. Fix $k \in \mathbb{N}$ and choose

$$P_{\max} \in \arg\max_{P \in \mathcal{A}^k} \rho(P)^{1/k}.$$

For the left-hand side we obtain, for any $r \in \mathbb{N}$,

$$\begin{aligned}
\max_{P \in \mathcal{A}^k} \rho(P)^{1/k} &= \rho(P_{\max})^{1/k} \\
&= \rho(P_{\max})^{r/(kr)} \\
&= \rho(P_{\max}^r)^{1/(kr)} \\
&\leq \max_{P \in \mathcal{A}^{kr}} \rho(P)^{1/(kr)} \\
&\leq \limsup_{r \to \infty} \max_{P \in \mathcal{A}^{kr}} \rho(P)^{1/(kr)} \\
&\leq \limsup_{n \to \infty} \max_{P \in \mathcal{A}^n} \rho(P)^{1/n} \\
&= \mathrm{JSR}(\mathcal{A}).
\end{aligned} \tag{1.6}$$

For the right-hand side, set

$$M_n := \max_{P \in \mathcal{A}^n} \|P\| \qquad (n \in \mathbb{N}).$$

By submultiplicativity of the norm, every $Q \in \mathcal{A}^{m+n}$ can be written as $Q = RS$ with $R \in \mathcal{A}^m$ and $S \in \mathcal{A}^n$, hence

$$\|Q\| \leq \|R\| \, \|S\| \leq M_m M_n, \tag{1.7}$$

and taking maxima over $Q \in \mathcal{A}^{m+n}$ yields

$$M_{m+n} \leq M_m M_n \qquad \text{for all } m, n \in \mathbb{N}. \tag{1.8}$$

Define the real sequence $a_n := \log M_n$. Then from (1.8),

$$a_{m+n} = \log M_{m+n} \leq \log(M_m M_n) = \log M_m + \log M_n = a_m + a_n, \tag{1.9}$$

so $(a_n)_{n \geq 1}$ is subadditive. By Fekete's lemma, the limit $\lim_{n \to \infty} a_n/n$ exists and equals $\inf_{n \geq 1} a_n/n$. In particular, for the fixed k we have

$$\lim_{n \to \infty} \frac{a_n}{n} = \inf_{n \geq 1} \frac{a_n}{n} \leq \frac{a_k}{k}. \tag{1.10}$$

Exponentiating (1.10) gives

$$\begin{aligned}\lim_{n\to\infty} M_n^{1/n} &= \exp\left(\lim_{n\to\infty} \frac{a_n}{n}\right) \\ &\leq \exp\left(\frac{a_k}{k}\right) \\ &= \exp\left(\frac{\log M_k}{k}\right) \\ &= M_k^{1/k} = \max_{P\in\mathcal{A}^k} \|P\|^{1/k}.\end{aligned} \tag{1.11}$$

Since $\mathrm{JSR}(\mathcal{A}) = \lim_{n\to\infty} \max_{P\in\mathcal{A}^n} \|P\|^{1/n} = \lim_{n\to\infty} M_n^{1/n}$ by definition, the right-hand inequality in 1.5 follows. □

This inequality underlies many computational methods, since it yields computable lower and upper bounds on $\mathrm{JSR}(\mathcal{A})$. As $k \to \infty$, the upper bound $\max_{P\in\mathcal{A}^k} \|P\|^{1/k}$ converges to $\mathrm{JSR}(\mathcal{A})$. The lower bounds $\max_{P\in\mathcal{A}^k} \rho(P)^{1/k}$ need not converge, but their limit superior equals $\mathrm{JSR}(\mathcal{A})$ by Proposition 1.3.4.

Similarity Invariance and Reducibility

Proposition 1.3.6. *(Jungers, 2009) The* JSR *is invariant under similarity transformations, i.e., for any invertible matrix T we have*

$$\mathrm{JSR}(\mathcal{A}) = \mathrm{JSR}(T^{-1}\mathcal{A}T). \tag{1.12}$$

Proof. This follows from the definition of the JSR and the properties of the spectral radius.

$$\begin{aligned}\mathrm{JSR}(T^{-1}\mathcal{A}T) &= \limsup_{k\to\infty} \max_{A_i\in\mathcal{A}^k} \rho(T^{-1}A_{i_k}T \cdot \ldots \cdot T^{-1}A_{i_1}T)^{1/k} \\ &= \limsup_{k\to\infty} \max_{A_i\in\mathcal{A}^k} \rho(T^{-1}A_{i_k} \cdot \ldots \cdot A_{i_1}T)^{1/k} \\ &= \limsup_{k\to\infty} \max_{A_i\in\mathcal{A}^k} \rho(A_{i_k} \cdot \ldots \cdot A_{i_1})^{1/k} \\ &= \mathrm{JSR}(\mathcal{A})\end{aligned}$$

□

Definition 1.3.7. (Jungers, 2009) A set of matrices is called *(commonly) reducible* if there exists a nontrivial subspace of $\mathbb{R}^n$ that is invariant under all matrices in the set. This means there exists a change of basis that block-triangularizes all matrices in $\mathcal{A}$ at the same time. If $\mathcal{A}$ is not reducible, it is called *irreducible*.

Proposition 1.3.8. *(Protasov, 1997) If $\mathcal{A}$ is reducible, then*

$$\mathrm{JSR}(\mathcal{A}) = \max\{\mathrm{JSR}(\mathcal{B}), \mathrm{JSR}(\mathcal{D})\}, \tag{1.13}$$

where $\mathcal{B} = \{B_1, \ldots, B_n\}$ and $\mathcal{D} = \{D_1, \ldots, D_n\}$ are the blocks of the block-triangularized matrices

$$T^{-1}A_jT = \begin{bmatrix} B_j & C_j \\ 0 & D_j \end{bmatrix}$$

under some invertible matrix T.

Example 1.3.9.

$$A_1 = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \quad A_2 = \begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix}$$

These matrices are reducible because there exists a nontrivial change of basis that transforms them into block-triangular form. The eigenvectors common to both matrices are

$$v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Using the change of basis matrix

$$T = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix},$$

we compute the similarity transformations

$$T^{-1}A_1T = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}, \quad T^{-1}A_2T = \begin{bmatrix} 5 & 0 \\ 0 & 1 \end{bmatrix}.$$

Since both matrices are upper triangular in this basis, they share a common invariant subspace, proving that they are reducible. Now, from Proposition 1.3.8, we know that the JSR can be calculated as

$$\mathrm{JSR}(\mathcal{A}) = \max\{\mathrm{JSR}(\{3, 5\}), \mathrm{JSR}(\{1, 1\})\} = 5$$

This can be applied iteratively until the sets of blocks are all irreducible. The problem is thereby split into similar problems of smaller dimension. For the following considerations we can now assume $\mathcal{A}$ to be irreducible.

Finiteness Property and Spectrum-Maximizing Products

Definition 1.3.10. (Lagarias and Wang, 1995) A matrix set $\mathcal{A} = \{A_1, \ldots, A_n\}$ is said to possess the *finiteness property* if there exists a finite sequence of matrices $A_{i_1}, \ldots, A_{i_k}$ such that

$$\mathrm{JSR}(\mathcal{A}) = \|A_{i_k} \cdots A_{i_1}\|^{1/k}. \tag{1.14}$$

Any such product is referred to as a *spectrum-maximizing product* (s.m.p.).

Although the finiteness property does not hold for all matrix sets, it is crucial for the effectiveness and termination of most algorithmic approaches. This is shown, for example, in (Blondel et al., 2003).

Computational Complexity of JSR Computation

The computation of the joint spectral radius (JSR) is theoretically challenging due to several complexity results.

Approximating the JSR is NP-hard (Tsitsiklis and Blondel, 1997), meaning no polynomial-time algorithm is expected unless P = NP. Moreover, for general sets of matrices, even deciding whether the JSR is strictly less than one is undecidable (Blondel and Tsitsiklis, 2000). Despite these limitations, many structured instances still allow efficient numerical approximation.

S.m.p.-Candidates and Generators

Approximating the JSR, using the invariant-polytope or the finite-tree algorithms, requires identifying candidate products or *generators* of the matrix set that contribute most significantly to the asymptotic growth rate. These generators are often derived through optimization techniques and their identification is a key step in computational algorithms.

Spectral Gap

Definition 1.3.11. (Mejstrik and Reif, 2025, Definition 1.9) A finite matrix family $\mathcal{A}$ is said to possess a *spectral gap* if there exists a constant $\gamma \in [0, 1)$ such that for every $k \in \mathbb{N}$ and every product $A_I \in \mathcal{A}^k$,

$$\rho(A_I) = \mathrm{JSR}(\mathcal{A})^k \quad \text{or} \quad \rho(A_I) \leq \gamma \ \mathrm{JSR}(\mathcal{A})^k.$$

Equivalently, every non-s.m.p. product is uniformly separated from the maximal growth rate by the factor $\gamma < 1$.

Remark 1.3.12. (Mejstrik and Reif, 2025; Guglielmi and Protasov, 2024) In the invariant-polytope setting, a spectral gap is a key finite-termination assumption together with uniqueness (up to cyclic permutations and powers) and simplicity of the leading eigenvalue of the s.m.p. candidate. Later chapters show where this gap condition enters the algorithmic termination analysis. We also note that (Guglielmi and Protasov, 2024) studies a related notion of gap as separation between the first and second Lyapunov exponents for matrix families.

Example 1.3.13. Consider

$$A_1 = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}, \qquad A_2 = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}, \qquad \mathcal{A} = \{A_1, A_2\}.$$

Since $\max\{\rho(A_1), \rho(A_2)\} = 2$ and $\|A_1\|_\infty = \|A_2\|_\infty = 2$, we get $\mathrm{JSR}(\mathcal{A}) = 2$. Any product $A_I \in \mathcal{A}^k$ has the form

$$A_I = A_1^m A_2^{k-m} = \begin{bmatrix} 2^m & 0 \\ 0 & 2^{k-m} \end{bmatrix},$$

so

$$\rho(A_I) = 2^{\max\{m, k-m\}}.$$

If $m = 0$ or $m = k$, then $\rho(A_I) = 2^k = \mathrm{JSR}(\mathcal{A})^k$ and A_I is an s.m.p. product. If $1 \leq m \leq k-1$, then

$$\rho(A_I) \leq 2^{k-1} = \tfrac{1}{2}\ \mathrm{JSR}(\mathcal{A})^k.$$

Hence Definition 1.3.11 holds with $\gamma = \frac{1}{2}$.

1.4. Preprocessing and Normalization

Both algorithms use the same preprocessing pipeline:

We want to find the JSR of the finite set of matrices $\mathcal{A} = \{A_1, \cdots, A_n\}$.

1. **Assumptions**: $\mathcal{A}$ is irreducible.
2. **Candidates**: Efficiently find products $P = A_{i_k} \cdots A_{i_1}$ of matrices from $\mathcal{A}$ that maximize the averaged spectral radius $\lambda := \rho(P)^{\frac{1}{k}}$.
3. **Rescaling**: Transform $\mathcal{A} \to \tilde{\mathcal{A}}$ with $\tilde{A}_i := \frac{1}{\lambda} A_i$.
4. **Verification**: Establish that $\mathrm{JSR}(\tilde{\mathcal{A}}) = 1$ using the three-member inequality. By homogeneity, this is equivalent to $\mathrm{JSR}(\mathcal{A}) = \lambda$.

The considered algorithms differ only in step 4. The invariant-polytope algorithm tries to find a vector norm whose induced matrix norm already bounds the matrices from $\mathcal{A}$ sufficiently. The finite-tree algorithm, on the other hand, bounds products using a partitioning space that separates each product into terms bounded by 1 and a remainder term that does not grow fast enough to overcome the k-th root in the JSR definition (polynomial growth).

2. Invariant Polytope Algorithm

In this chapter, we focus on the invariant-polytope algorithm. At its core, this algorithm seeks a polytope $\mathcal{P} \subset \mathbb{R}^d$ that is invariant under the action of the already scaled matrices in $\tilde{\mathcal{A}}$. The corresponding Minkowski functional of this polytope is then a so-called extremal norm for the family $\mathcal{A}$. As we will see, this is enough to prove that the chosen candidate is indeed an s.m.p., which is our goal.

2.1. Extremal and Invariant Norms

Definition 2.1.1. (Barabanov, 1988) For a family of matrices $\mathcal{A}$ a norm $\|\cdot\|$ is called *invariant* or *Barabanov* if there exists a number $\lambda \geq 0$ such that

$$\max_{A \in \mathcal{A}} \|Ax\| = \lambda \|x\|, \quad \text{for all } x \in \mathbb{R}^d.$$

The scalar in Definition 2.1.1 equals $\mathrm{JSR}(\mathcal{A})$.

The next theorem is needed because it guarantees that invariant norms exist under irreducibility.

Theorem 2.1.2. *(Barabanov, 1988) Every irreducible family $\mathcal{A}$ possesses an invariant norm.*

We are interested in norms with a weaker condition.

Definition 2.1.3. (Protasov, 1996) A norm $\|\cdot\|$ is called *extremal* for a family of matrices $\mathcal{A}$ if

$$\max_{A \in \mathcal{A}} \|Ax\| \leq \mathrm{JSR}(\mathcal{A}) \|x\| \quad \text{for all } x \in \mathbb{R}^d.$$

Every invariant norm is, of course, extremal, which guarantees the existence of such a norm.

Theorem 2.1.4. *For every extremal norm $\|\cdot\|$ of a family of matrices $\mathcal{A}$*

$$\max_{P\in\mathcal{A}^k}\|P\| = \mathrm{JSR}(\mathcal{A})^k$$

holds for every $k \in \mathbb{N}$.

Proof. By submultiplicativity and extremality of the norm, we have

$$\begin{aligned}\max_{P\in\mathcal{A}^k}\|P\| &= \max_{d_k,\dots,d_1}\|A_{d_k}\cdots A_{d_1}\| \\ &\leq \max_{d_k,\dots,d_1}\|A_{d_k}\|\cdots\|A_{d_1}\| \\ &\leq \mathrm{JSR}(\mathcal{A})^k\end{aligned}$$

By the three-member inequality (1.5), we know

$$\begin{aligned}\mathrm{JSR}(\mathcal{A}) &\leq \max_{P\in\mathcal{A}^k}\|P\|^{\frac{1}{k}} \\ \mathrm{JSR}(\mathcal{A})^k &\leq \max_{P\in\mathcal{A}^k}\|P\|\end{aligned}$$

for every $k \in \mathbb{N}$.

□

Remark 2.1.5. In particular, when $k = 1$, we have $\max_{A\in\mathcal{A}}\|A\| = \mathrm{JSR}(\mathcal{A})$. The existence of an extremal norm for $\mathcal{A}$ therefore enables the exact computation of the joint spectral radius.

In the following, we build an extremal norm whose unit ball is a balanced polytope, as introduced in (Protasov, 1996) and (Guglielmi and Zennaro, 2008).

Definition 2.1.6. A polytope $\mathcal{P}$ is called *balanced* if it spans the whole space and is equal to the symmetrized convex hull of a finite set of vertices.

$$\mathcal{P} = \mathrm{co_s}(V) := \mathrm{co}(V, -V) = \{x \in \mathbb{R}^d : x = \sum_{i=1}^{M}\alpha_i v_i \text{ and } \sum_{i=1}^{M}|\alpha_i| \leq 1\}$$

If a polytope satisfies $\mathcal{A}\mathcal{P} \subseteq \mathcal{P}$ for some family $\mathcal{A}$, it is called *invariant under* $\mathcal{A}$.

Definition 2.1.7. (Magaril-Il'yaev and Tikhomirov, 2001) Let $C \subseteq \mathbb{R}^d$ be a nonempty, convex set such that $0 \in \mathrm{int}(C)$. The *Minkowski functional* (or *gauge function*) associated to C is the mapping $p_C : \mathbb{R}^d \to [0, \infty)$ defined by:

$$p_C(x) = \inf\{\lambda > 0 \mid x \in \lambda C\} \quad \text{for all } x \in \mathbb{R}^d.$$

Remark 2.1.8. In general, the Minkowski functional is only a seminorm on $\mathbb{R}^d$. If $\mathcal{P}$ is a balanced polytope, then the associated Minkowski functional defines a norm on $\mathbb{R}^d$. Moreover, every induced matrix norm on $\mathbb{R}^{d\times d}$ is submultiplicative. In both cases, we denote the norm by $\|\cdot\|_{\mathcal{P}}$.

Theorem 2.1.9. *(Guglielmi and Protasov, 2013) If a balanced polytope $\mathcal{P}$ is invariant under $\tilde{\mathcal{A}}$ from Section 1.4, the corresponding Minkowski functional is an extremal norm for the family $\mathcal{A}$ and* $\mathrm{JSR}(\mathcal{A}) \leq \hat{\rho}$.

Proof.

$$\begin{aligned}\tilde{\mathcal{A}}\mathcal{P} \subseteq \mathcal{P} &\implies \mathcal{A}\mathcal{P} \subseteq \hat{\rho}\mathcal{P} \\ &\implies \|A\|_{\mathcal{P}} \leq \hat{\rho} \quad \text{for all } A \in \mathcal{A} \\ &\implies \max_{A\in\mathcal{A}} \|A\|_{\mathcal{P}} \leq \hat{\rho}\end{aligned}$$

□

By the previous theorem, we only need to find a balanced polytope that is invariant under the action of $\tilde{\mathcal{A}}$. Then we have proved that the chosen candidate from the preprocessing is indeed an s.m.p.

2.2. Algorithmic Structure

After preprocessing has been carried out, we end up with a candidate $\tilde{\Pi} = \tilde{A}_{d_k} \cdots \tilde{A}_{d_1}$ for our finiteness-property hypothesis. We want to establish $\mathrm{JSR}(\tilde{\mathcal{A}}) = 1$ by finding an invariant balanced polytope $\mathcal{P} = \mathrm{co_s}(V)$. We define $V := \{v_1, \cdots, v_k\}$, where v_1 is the leading eigenvector of $\tilde{\Pi}$ (assumed real and unique) and $v_i := \tilde{A}_{d_{i-1}} \cdots \tilde{A}_{d_1} v_1$ are the leading eigenvectors of the cyclic permutations of $\tilde{\Pi}$. Now we add new vertices to V iteratively by multiplying with matrices from $\tilde{\mathcal{A}}$, i.e.

$$V \overset{\cup}{\leftarrow} \tilde{\mathcal{A}}V :\iff V \leftarrow V \cup \tilde{\mathcal{A}}V.$$

All vertices that fall into the symmetrized convex hull of V can be disregarded, since they do not contribute to a change of the polytope $\mathcal{P}$ and therefore reduce computational effort. In practice, it is very important to drop as many vertices as possible, since V grows exponentially. This is done by solving a standard linear programming problem to determine whether $\|Av\|_{\mathrm{co_s}(V)} \geq 1$. If the algorithm does not produce new vertices that lie outside the current polytope for every $\tilde{A} \in \tilde{\mathcal{A}}$, then it is invariant by definition, and the finiteness property has been proved for the chosen candidate.

Algorithm 1 Invariant-polytope algorithm

```
V := {v_1, ··· , v_M}
V_new ← V
while V_new ≠ ∅ do
    V_rem ← V_new
    V_new ← ∅
    for v ∈ V_rem do
        for A ∈ 𝒜 do
            if ‖Av‖_{co_s(V)} ≥ 1 then
                V ←∪ Av
                V_new ←∪ Av
return co_s(V)
```

2.3. Stopping Criterion and Candidate Rejection

The runtime of Algorithm 1 can be high in practice, so it is important to detect incorrect candidates early and control the growth of the vertex set V. A common practical safeguard is to set an upper bound on the number of vertices. For rejecting an incorrect s.m.p. candidate, (Guglielmi and Protasov, 2013) proposed the following method. In the case where the candidate $\tilde{\Pi}$ has a unique simple leading eigenvalue and there exists only one s.m.p., define v_1^* as the leading eigenvector of $\tilde{\Pi}^*$ (the conjugate operator of $\tilde{\Pi}$), normalized by $(v_1, v_1^*) = 1$, and define $v_i^* := \tilde{A}_{d_i}^* \cdots \tilde{A}_{d_k}^* v_1^*$, which are the leading eigenvectors of the conjugates of the cyclic permutations of $\Pi \in \mathcal{A}^k$. If $\|Av\|_{\mathrm{co_s}(V)} > 1$ and there exists a j such that $|(v_j^*, Av)| > 1$, then the chosen candidate is not an s.m.p. and must be rejected.

2.4. Termination Conditions

There are rare cases where the finiteness property does not hold, or where no invariant polytopes exist. In those cases, the algorithm will not find the value of the JSR.

2.5. Rebalancing and Additional Initial Vertices

Three years after publishing the fundamental algorithm with the stopping criterion, the authors released a new paper on rebalancing multiple s.m.p.s, as well as starting with extra vertices so that the polytope is better conditioned (Guglielmi and Protasov, 2016). In that setting, termination with multiple s.m.p.s can be achieved, but it is substantially more delicate than in the unique-s.m.p. case.

2.6. Complex Variants

If the leading eigenvector associated with a candidate product is complex, then the real balanced-polytope norm is no longer sufficient and must be replaced by a *complex-balanced polytope norm*. This changes the convex-hull membership tests in the linear-programming step, while the high-level structure of the algorithm remains the same.

In the present thesis, we restrict attention to the real case. For details on complex variants, we refer to (Guglielmi et al., 2005; Guglielmi and Zennaro, 2007; Guglielmi and Protasov, 2013).

3. Finite-Tree Algorithm

In this chapter we introduce definitions and concepts for another approach to JSR computation. The *finite-tree algorithm* decomposes arbitrary products $P \in \mathcal{A}^k$ into subproducts that are 1-bounded with respect to a chosen norm.

3.1. Tree-Based Notation and Definitions

Definition 3.1.1. A *tree* is a connected, acyclic graph T with a distinguished node t_0 called the *root*. The tree structure induces a natural hierarchy: for any node t_i, there exists a unique path from t_0 to t_i. The tree can then be described in terms of the following concepts:

- **Parent**: For any node $t \in T$, the node preceding t on the path from the root t_0 is called its *parent*.
- **Children**: The nodes adjacent to a given node t that are further from the root t_0 are called its *children*.
- **Leaf**: A node with no children is called a *leaf*.
- **Internal vertex**: A node with at least one child is called an *internal* or *non-leaf* node.

We define $T_{\mathcal{A}}$ as the tree consisting of all products in the set $\bigcup_{k \in \mathbb{N}_0} \mathcal{A}^k$ where every multi-index represents a node. For the empty index ($k = 0$) we set $A_{[]} = I_d$, the identity matrix in $\mathbb{R}^{d \times d}$, and call this node the root. An example can be seen in Figure 3.1.

The objective of the finite-tree algorithm is to find a sub-tree $T \subset T_{\mathcal{A}}$ such that every leaf of T has norm less than 1 and every node that is not a leaf has all positive children $A_1, \ldots, A_n$. If such a tree is found, every product $P = A_{[i_1,\ldots,i_k]} \in \mathcal{A}^k$ can be decomposed into factors encoded by leaves and one possible remainder term encoded by an internal node. This suffices to prove $\mathrm{JSR}(\tilde{\mathcal{A}}) = 1$.

In general, such trees are not finite and are therefore not directly usable for computation. In (Möller and Reif, 2014), the authors introduce an encoding with generators and set-valued nodes to obtain a tractable search procedure.

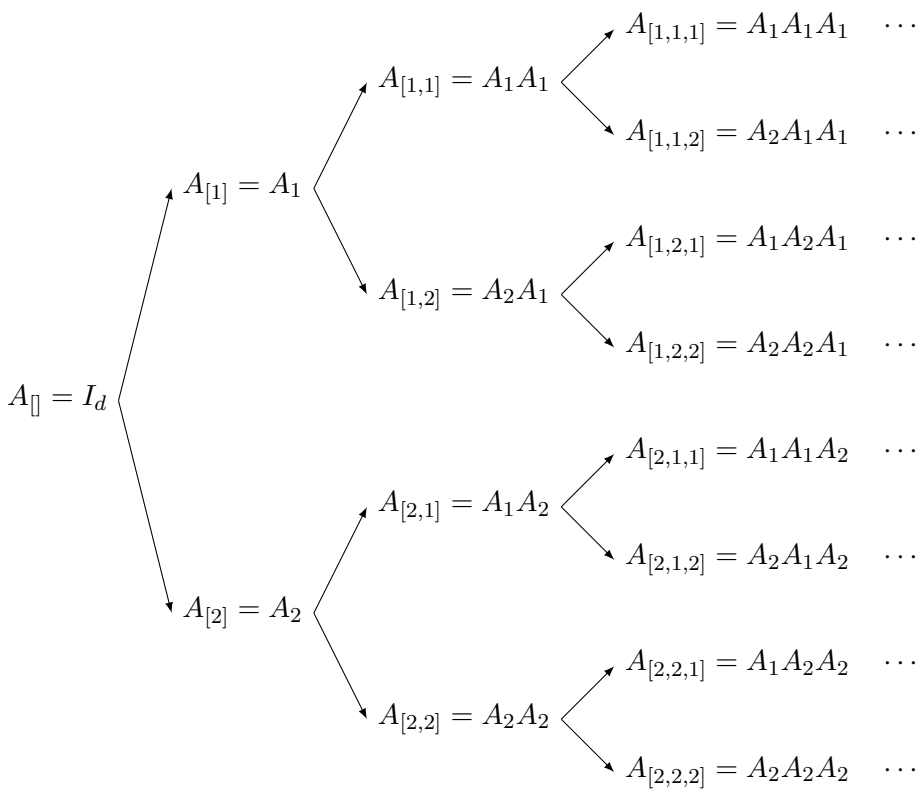

Figure 3.1.: An example of the tree $T_{\mathcal{A}}$ on $\bigcup_{k\in\mathbb{N}_0} \mathcal{A}^k$ with $\mathcal{A} = \{A_1, A_2\}$.

We fix a set $\mathcal{G} = \{G_1, \cdots, G_m\}$ with $G_i \in \mathcal{A}^{k_i}$ and $\rho(G_i)^{\frac{1}{k_i}} \leq 1$. All products from this set will be called *generators*. We also define $A_l := G_{-l}$ for $l \in \{-m, \ldots, -1\}$.

Definition 3.1.2. (Möller and Reif, 2014)

The set $\mathcal{J}^k := \{-m, \ldots, -1, 1, \ldots, n\}^k$ denotes the collection of all multi-indices of length k with entries in $\{-m, \ldots, -1, 1, \ldots, n\}$. For a multi-index $J = [j_1, \ldots, j_k] \in \mathcal{J}^k$, the corresponding matrix product set is defined by:

$$A_J := \{A_{j_k}^{e_k} \cdots A_{j_1}^{e_1} \mid e_i = 1 \text{ if } j_i > 0 \text{ or } e_i \in \mathbb{N}_0 \text{ otherwise}\}$$

Again, this creates a tree structure in a similar way to positive multi-indeces, except now the nodes represent (possibly infinite) sets of products.

Definition 3.1.3. Let $J = [j_1, \cdots, j_k]$ be a multi-index and let $j_{k+1} \in \mathcal{J}$. We define the extension

$$[J, j_{k+1}] := [j_1, \cdots, j_k, j_{k+1}].$$

Nested concatenations are always flattened a priori.

Definition 3.1.4. Let $t \in T$ be a node with the multi-index J. A child node $\tilde{t}$ with the multi-index $[J, j_{k+1}]$ is called:

- *positive* if j_{k+1} is positive.
- *generator* if j_{k+1} is negative.

Now we want to find a subtree of a special form.

Definition 3.1.5. (Möller and Reif, 2014) A so-called $(\mathcal{A}, \mathcal{G})$-tree T has the following structure:

- The root node contains the identity matrix: $t_0 = \{I_d\}$.
- For every node $t \in T \setminus \{t_0\}$, there is exactly one parent node in T.
- Each node $t \in T$ is either:
  - A leaf,
  - A parent of exactly n positive children.
  - A parent with at least one generator child.

Definition 3.1.6. (Möller and Reif, 2014) A node in the tree is called *covered* if it is a subset of one of its ancestors in the tree. Otherwise, it is called *uncovered*. The set of uncovered leaves is denoted as

$$\mathcal{L}(T) := \{t \in T : t \text{ is an uncovered leaf}\}$$

and called *leafage* of the tree T.

Definition 3.1.7. (Möller and Reif, 2014) An $(\mathcal{A}, \mathcal{G})$-tree T is called *1-bounded* with respect to a matrix norm $\|\cdot\|$ if

$$\sup_{L \in \mathcal{L}(T)} \|L\| \leq 1.$$

If $\sup_{L \in \mathcal{L}(T)} \|L\| < 1$, then the tree is called *strictly 1-bounded*.

3.2. Algorithmic Structure

The finite-tree algorithm starts with the root node $t_0 = \{I_d\}$ and performs a non-exhaustive depth-first search in the space of $(\mathcal{A}, \mathcal{G})$-trees. In each iteration, it checks whether every active leaf satisfies

$$\sup\{\|P\| : P \in L\} \leq 1.$$

If this fails for a leaf, the procedure SELECTCHILDREN chooses either all positive children or a selected subset of generator children.

Algorithm 2 Finite-tree algorithm

Input: $\mathcal{A} = \{A_1, \ldots, A_n\}$ and $\mathcal{G} = \{G_1, \cdots, G_m\}$
Output: If the algorithm terminates, the tree condition holds
$Q \leftarrow \{[1], \cdots, [n]\}$
while $Q \neq \emptyset$ **do**
  $Q_{\text{new}} \leftarrow \emptyset$
  for $J \in Q$ **do**
    if $\sup\{\|P\| : P \in A_J\} > 1$ **then**
      $\mathcal{C} \leftarrow \textsc{selectChildren}(J, \mathcal{A}, \mathcal{G})$
      $Q_{\text{new}} \overset{\cup}{\leftarrow} \{[J, j] : j \in \mathcal{C}\}$
  $Q \leftarrow Q_{\text{new}}$

Remark 3.2.1. (Möller and Reif, 2014) The procedure SELECTCHILDREN determines whether all positive children (from 1 to n) or a subset of generator children (from -1 to $-m$) is chosen, according to the theory established there.

Lemma 3.2.2. *For every $(\mathcal{A}, \mathcal{G})$-tree T there exists a strictly positive, monotone polynomial p such that for every node $t \in T$ and every product $P \in \mathcal{A}^k$ encoded by t, $\|P\| \leq p(k)$ holds.*

Proof. Take an arbitrary node $t \in T$ with multi-index $J = [j_1, \cdots, j_l]$. Now every j_i encodes either a matrix from $\mathcal{A}$ or arbitrary powers of a generator from $\mathcal{G}$. Since all those matrices have spectral radius less than or equal to 1, their Jordan normal forms grow at most polynomially and thus, by norm equivalence in finite-dimensional vector spaces, there exists a strictly positive, monotone polynomial p_i with

$$\|A_{j_i}^e\| \leq p_i(e) \quad \text{for all } e \in \mathbb{N}_0.$$

We define:

$$p_t := p_l \cdots p_1$$
$$p := \sum_{t \in T} p_t$$

The sum over all of T is finite since T has only finitely many nodes. Take now an arbitrary product $P = A_{j_l}^{e_l} \cdots A_{j_1}^{e_1} \in \mathcal{A}^k$ that is encoded by t.

Now we have:

$$\begin{aligned}
\|P\| &= \|A_{j_l}^{e_l} \cdots A_{j_1}^{e_1}\| \\
&\leq \|A_{j_l}^{e_l}\| \cdots \|A_{j_1}^{e_1}\| \\
&\leq p_l(e_l) \cdots p_1(e_1) \\
&\leq p_l(k) \cdots p_1(k) \\
&= p_t(k) \leq p(k)
\end{aligned}$$

Since the node t and the product P were chosen arbitrary, this concludes the proof. □

Theorem 3.2.3. *(Möller and Reif, 2014) If a 1-bounded $(\mathcal{A}, \mathcal{G})$-tree is found for a given matrix set $\mathcal{A}$ and generator set $\mathcal{G}$, then* $\mathrm{JSR}(\mathcal{A}) = 1$.

Proof. We take an arbitrary positive multi-index $I = [i_1, \cdots, i_k]$. Our found $(\mathcal{A}, \mathcal{G})$-tree T allows for a unique decomposition of this multi-index into $I_k, \cdots, I_1$ where

$$A_I = A_{I_k} A_{I_{k-1}} \cdots A_{I_1}.$$

Each sub-multi-index $I_j \quad j = 1, \cdots, k-1$ encodes either a leaf of T or is the empty set and I_k encodes any node of T. Now due to Lemma 3.2.2 there exists a polynomial p such that

$$\|A_I\| = \|A_{I_k} A_{I_{k-1}} \cdots A_{I_1}\| \leq \|A_{I_k}\| \cdot \|A_{I_{k-1}}\| \cdots \|A_{I_1}\| \leq \|A_{I_k}\| \leq p(k).$$

The last equality holds since all $I_j \quad j = 1, \cdots, k-1$ are either the empty multi-index or leafs such that their products A_{I_j} are bounded by 1. Furthermore A_{I_k} can be bounded by the polynomial p and the length of the product, which is less than k. Now we have

$$\mathrm{JSR}(\mathcal{A}) = \lim_{k \to \infty} \max_{P \in \mathcal{A}^k} \|P\|^{\frac{1}{k}} \leq \lim_{k \to \infty} p(k)^{\frac{1}{k}} = 1$$

since the chosen multi-index was arbitrary. □

Remark 3.2.4. (Möller and Reif, 2014) The decomposition used in the proof of Theorem 3.2.3 exists and is unique due to the structure of an $(\mathcal{A}, \mathcal{G})$-tree.

3.3. Termination Conditions

Remark 3.3.1. (Möller and Reif, 2014) In the cited work, the authors show that the use of generators is essential for ensuring termination in cases where the original invariant-polytope algorithm fails to terminate. Deciding when to use generators instead of all positive children is still part of active research and is handled by heuristics. In particular, termination with multiple s.m.p. candidates is substantially more delicate.

4. Hybrid Polytope-Tree Approach

In this chapter we discuss two hybridization strategies for the previously introduced algorithms and then present the main result of this thesis: the tree-flavored invariant-polytope algorithm with its termination result.

4.1. Hybridization Strategies

At its core, the invariant-polytope algorithm constructs a polytope norm adapted to the matrix family, while the finite-tree algorithm controls growth through product decompositions. This naturally suggests a hybridization.

One option is modular: first run the invariant-polytope algorithm for several iterations, then use the resulting norm inside the finite-tree phase. A second option is concurrent: update norm information and tree-decomposition information in each iteration. Following (Mejstrik and Reif, 2025), we use the concurrent variant.

4.2. Problem Formulation

After preprocessing from Section 1.4, we obtain a scaled family $\tilde{\mathcal{A}}$. From the chosen candidate and the three-member inequality (1.5), we already know $\mathrm{JSR}(\tilde{\mathcal{A}}) \geq 1$. Hence proving $\mathrm{JSR}(\tilde{\mathcal{A}}) \leq 1$ is sufficient.

To do this, we show that there exist a polynomial p and a norm $\|\cdot\|$ such that, similarly to Chapter 3, every product can be bounded polynomially in its length:

$$\|P\| \leq p(k) \quad \text{for all } P \in \tilde{\mathcal{A}}^k,\ k \in \mathbb{N}.$$

The polynomial bound is obtained by decomposing each multi-index into finitely many 1-bounded parts plus one residual part controlled by finite-tree arguments.

4.3. Hybrid Algorithm

The algorithm starts, as in the invariant-polytope method, from a set V_0 of starting vertices. Each vertex set V_r defines a vector polytope norm $\|\cdot\|_{V_r}$. At iteration r, for every vertex $v \in V_r$, we build a test $(\mathcal{A}, \mathbf{G})$-tree T_v. If some L in the leafage of T_v satisfies $\|Lv\|_{V_r} \geq 1$, then Lv is added as a new vertex candidate. If all test-tree leafages in one iteration are 1-bounded by the current polytope norm, the algorithm terminates.

Algorithm 3 Invariant-tree-algorithm

Input: $\mathcal{A}$ (set of matrices), $\mathbf{G}$ (set of generators), V (set of starting vertices)
$V_{\text{new}} \leftarrow V$
while $V_{\text{new}} \neq \emptyset$ **do**
  $V_{\text{rem}} \leftarrow V_{\text{new}}$
  $V_{\text{new}} \leftarrow \emptyset$
  for $v \in V_{\text{rem}}$ **do**
    Construct minimal $(\mathcal{A}, \mathbf{G})$-tree $\mathbf{T}$ as discussed in Section 4.6
    $\mathcal{W} \leftarrow \{Lv : L \in \mathcal{L}(\mathbf{T}) \text{ and } \|Lv\|_{\text{cos}(V)} \geq 1\}$
    if $\mathcal{W} \neq \emptyset$ **then**
      $V \overset{\cup}{\leftarrow} \mathcal{W}$
      $V_{\text{new}} \overset{\cup}{\leftarrow} \mathcal{W}$

4.4. Termination Theory

Theorem 4.4.1. *If Algorithm 3 terminates, then* $\text{JSR}(\mathcal{A}) \leq 1$.

4.4.1. Proof Strategy

Products are bounded by norms built during intermediate iterations, and submultiplicativity allows us to pass to the final norm. Therefore, we work with $\|\cdot\|_V$ for the final vertex set $V = \cup_{i=1}^{K} V_i$, where K is the final iteration. The proof generalizes the finite-tree argument by repeatedly expanding linear combinations and controlling their coefficients via 1-boundedness.

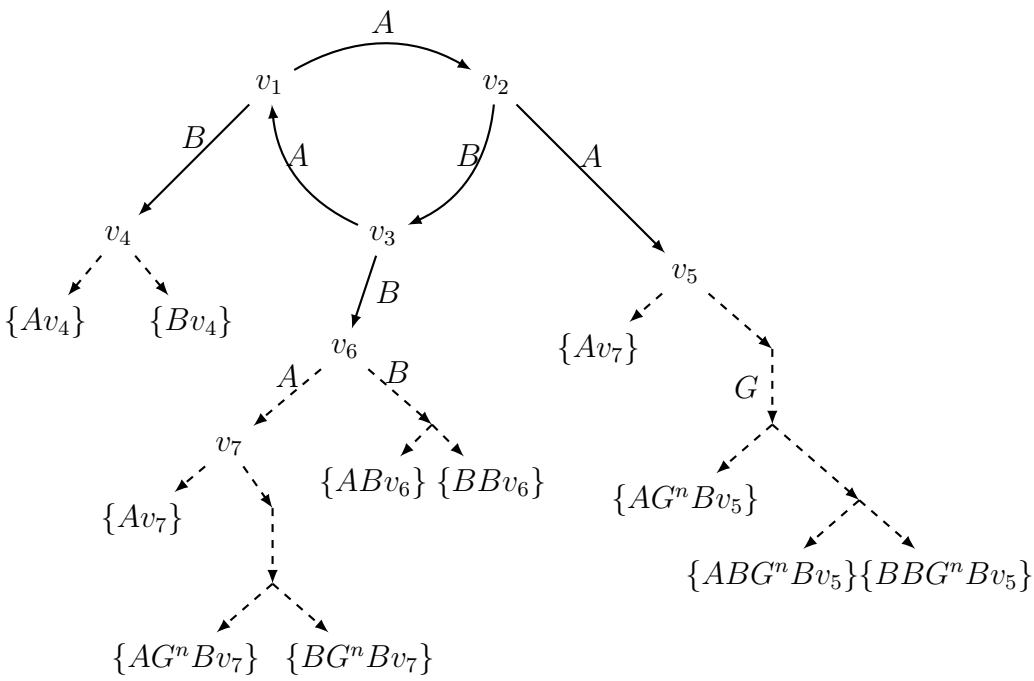

Figure 4.1.: Cyclic tree structure generated by the algorithm. Here $\mathcal{A}$ consists of two matrices A and B and the chosen candidate is $\Pi = ABA$. Dashed arrows indicate $(\mathcal{A}, \mathbf{G})$-trees used for bounds. Solid arrows indicate vertices added to V in one iteration. Leaf labels with powers G^n represent 1-bounded matrix-vector products for all $n \in \mathbb{N}_0$.

4.4.2. Auxiliary Termination Results

Lemma 4.4.2. *If Algorithm 3 terminates, there exists a monotone polynomial p such that for every product $P \in \mathcal{A}^k$ and $v \in V$, $\|Pv\|_{co_s(V)} \leq p(k)$ holds.*

Proof. If the algorithm terminates, then the set of vertices $V = \{v_1, \cdots, v_m\}$ is finite and for each $v \in V$ there exists an $(\mathcal{A}, \mathbf{G})$-tree T_v such that $\|Lv\|_{\mathrm{co_s}(V)} \leq 1$ for every $L \in \mathcal{L}(T_v)$, which means there exists $\lambda_1, \cdots, \lambda_m$ such that $Lv = \sum \lambda_j v_j$ with $\sum |\lambda_j| \leq 1$. According to Lemma 3.2.2, there exists a polynomial p_v such that for every product $P \in \mathcal{A}^k$ encoded by T_v, $\|Pv\|_{\mathrm{co_s}(V)} \leq p_v(k)$. Take an arbitrary product $P \in \mathcal{A}^k$ and $v \in V$. Define:

$$
\begin{aligned}
p &:= \sum_{v \in V} p_v \\
P'L &:= P \text{ with } L \in \mathcal{L}(T_v) \text{ if } P \notin T_v \text{ and } L := I \text{ else} \\
P'_{i_1 \ldots i_r} L_{i_1 \ldots i_r} &:= P'_{i_1 \ldots i_{r-1}} \text{ with } L_{i_1 \ldots i_r} \in \mathcal{L}(T_{v_{i_r}}) \text{ if } P'_{i_1 \ldots i_{r-1}} \notin T_{v_{i_r}} \\
\text{and } L_{i_1 \ldots i_r} &:= I \text{ else.} \\
\text{Where } i_j \in \mathcal{I} &:= \{1, \cdots, m\}
\end{aligned}
$$

These decompositions exist and are unique because of the special structure of $(\mathcal{A}, \mathbf{G})$-trees; details can be found in (Möller and Reif, 2014).
Now we have:

$$
\begin{aligned}
\|Pv\|_{co_s(V)} &= \|P'Lv\|_{co_s(V)} \\
&= \|P' \sum_{i_1 \in \mathcal{I}} \lambda_{i_1} v_{i_1}\|_{co_s(V)} \\
&= \sum_{i_1 \in \mathcal{I}} |\lambda_{i_1}| \|P' v_{i_1}\|_{co_s(V)} \\
&= \sum_{i_1 \in \mathcal{I}} |\lambda_{i_1}| \|P'_{i_1} L_{i_1} v_{i_1}\|_{co_s(V)} \\
&= \sum_{i \in \mathcal{I}^2} |\lambda_{i_1}| |\lambda_{i_2}| \|P'_i v_{i_2}\|_{co_s(V)} \\
&\cdots \\
&= \sum_{i \in \mathcal{I}^k} |\lambda_{i_1}| \cdot \ldots \cdot |\lambda_{i_k}| \|P'_i v_{i_k}\|_{co_s(V)} \\
&\leq \sum_{i \in \mathcal{I}^k} |\lambda_{i_1}| \cdot \ldots \cdot |\lambda_{i_k}| \; p_{v_{i_k}}(k) \\
&\leq p(k) \sum_{i \in \mathcal{I}^k} |\lambda_{i_1}| \cdot \ldots \cdot |\lambda_{i_k}| \\
&= p(k) \sum_{i_1 \in \mathcal{I}} |\lambda_{i_1}| \sum_{i_2 \in \mathcal{I}} |\lambda_{i_2}| \ldots \sum_{i_k \in \mathcal{I}} |\lambda_{i_k}| \\
&\leq p(k) \cdot 1 \cdot \ldots \cdot 1 \\
&\leq p(k)
\end{aligned}
$$

Here the first inequality works since after k steps the prefix P_i' is guaranteed to be in the tree $T_{v_{i_k}}$, and thus the norm is bounded by $p_{v_{i_k}}(k)$, which is less than $p(k)$ by definition. The second-to-last inequality follows because the sums of absolute coefficients in the corresponding linear combinations are bounded by 1 due to 1-boundedness of the $(\mathcal{A}, \mathbf{G})$-trees. □

4.4.3. Main Termination Result

Proof. Due to the auxiliary lemmata, Theorem 4.4.1 is now straightforward. Lemma 4.4.2 implies that for every $P \in \mathcal{A}^k$ and $v \in V$, $\|Pv\|_{\mathrm{co_s}(V)} \leq p(k)$. In particular, this implies

$$\|P\|_{\mathrm{co_s}(V)} \leq p(k) \quad \text{for all } P \in \mathcal{A}^k.$$

Taking the definition of JSR gives

$$\mathrm{JSR}(\mathcal{A}) = \lim_{k\to\infty} \max_{P\in\mathcal{A}^k} \|P\|^{\frac{1}{k}} \leq \lim_{k\to\infty} p(k)^{\frac{1}{k}} = 1.$$

□

Corollary 4.4.3. *If the chosen trees are always $T_v = \mathcal{A}$, then Algorithm 3 reduces to the invariant-polytope algorithm 1. Hence, the hybrid approach generalizes the original method.*

4.5. Remarks

4.5.1. Eventual Polytope Invariance

A notable disadvantage of the hybrid algorithm is that the computed polytope $\mathrm{co_s}(V)$ is generally not invariant under $\mathcal{A}$. Therefore, the polytope alone is not a certificate for an upper bound on $\mathrm{JSR}(\mathcal{A})$; the corresponding tree information must also be reported.

4.6. Implementation Considerations

This section fixes the tree class used in Algorithm 3 and states the corresponding minimal-tree definition used in implementation.

Definition 4.6.1. An $(\mathcal{A}, \mathbf{G})$-tree with one generator branch at the root, one path from the root to the covered node, and all branches required to satisfy Definition 3.1.5, is called a *minimal tree*. This tree depends only on $\mathcal{A}$ and the chosen generator G. It is denoted by $T_{(\mathcal{A},\mathbf{G})}$ and, when $\mathcal{A}$ is clear from context, shortened to $T_{\mathbf{G}}$.

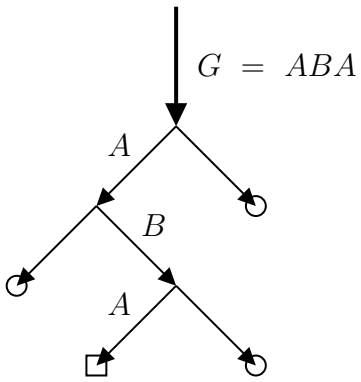

Figure 4.2.: Minimal tree structure $T_{(G,\mathcal{A})}$ for the candidate $G = ABA$ and $\mathcal{A} = \{A, B\}$. The arrows marked with circles are uncovered-leafs, the arrow marked with a square is a covered-leaf

5. Numerical Experiments

5.1. Runtime Analysis of Algorithm 3 on Generated Matrix Families

5.2. Comparison with Algorithm 1

5.3. Comparison with Algorithm 2

5.4. Case Studies on Previously Open Problems

6. Conclusions and Outlook

6.1. Summary of the Hybrid Framework

6.2. Resolved Problem Classes

6.3. Computational Efficiency

6.4. Remaining Open Problems

6.5. Directions for Future Research

6.5.1. Extension to the Complex Setting

6.5.2. Structured Nonnegative Matrix Families

6.5.3. Optimization of Tree Construction and Vertex Selection

A. Notation Glossary

A.1. Constants and Globally Defined Variables

A_i Matrix in the family $\mathcal{A}$.

d Dimension of each matrix $A_i \in \mathbb{R}^{d \times d}$.

n Number of matrices in $\mathcal{A}$.

$\mathcal{A} = \{A_1, \ldots, A_n\}$ Base set of matrices.

$\tilde{\mathcal{A}} = \{\tilde{A}_1, \ldots, \tilde{A}_n\}$ Scaled matrix set after preprocessing.

m Number of generators.

$\mathcal{G} = \{G_1, \ldots, G_m\}$ Set of generators.

k Length of a product.

α Scalar.

λ Eigenvalue.

x Vector.

ρ Spectral radius.

σ Singular value.

JSR Joint spectral radius operator.

A.2. Products and Multi-Indices

$\mathcal{I} = \{1, \ldots, n\}$ Positive matrix-index set.

$\mathcal{I}^k$ Set of positive multi-indices of length k.

i Positive index.

I Positive multi-index of length k, i.e. $I = [i_1, \ldots, i_k] \in \mathcal{I}^k$.

$A_I = A_{[i_1,\ldots,i_k]} = A_{i_k} \cdots A_{i_1}$ Product encoded by I.

$\mathcal{J} = \{-m, \dots, -1, 1, \dots, n\}$ Extended index set including generator indices.

$\mathcal{J}^k$ Set of extended multi-indices of length k.

j Extended index.

$\mathcal{A}_\mathcal{G}$ Family of primitive product sets:

$$\mathcal{A}_\mathcal{G} := \{\{G_m^r : r \in \mathbb{N}\}, \dots, \{G_1^r : r \in \mathbb{N}\}, \{A_1\}, \dots, \{A_n\}\}\,.$$

$A_j \in \mathcal{A}_\mathcal{G}$ Primitive set encoded by index j.

J Extended multi-index of length k, i.e. $J = [j_1, \dots, j_k] \in \mathcal{J}^k$.

$A_J = A_{[j_1,\dots,j_k]}$ Product set encoded by J.

$[J_1, j_3, J_4, [j_1, J_2]]$ Expressions are interpreted in flattened form.

A.3. Norms and JSR

$\|\cdot\|_{\mathrm{co_s}(V)} = \|\cdot\|_V$ Polytope norm notation used for matrices and vectors.

$\mathrm{JSR}(\mathcal{A}) := \lim\limits_{k\to\infty} \max\limits_{P\in\mathcal{A}^k} \|P\|^{1/k}$ Norm-based definition of JSR.

$\mathrm{JSR}(\mathcal{A}) = \limsup\limits_{k\to\infty} \max\limits_{P\in\mathcal{A}^k} \rho(P)^{1/k}$ Berger–Wang characterization.

A.4. Algorithmic Notation

Invariant-polytope algorithm Baseline algorithm from Chapter 2.

Finite-tree algorithm Tree-based algorithm from Chapter 3.

Invariant-tree algorithm Hybrid algorithm from Chapter 4.

M Number of vertices in a finite vertex set representation $V = \{v_1, \dots, v_M\}$.

V_k Set of vertices at step k.

R_k Set of vertices remaining to test at step k.

V Union of vertex sets over all steps.

$\overset{\cup}{\leftarrow}$ Appending by union in pseudocode.

A.5. Tree Terminology

T a tree is a directed graph.

Root Node without a predecessor.

Child and parent If a node $t \in T$ has predecessor $p \in T$, then t is a child of p and p is the parent of t.

Leaf Node without children.

Internal node Node with at least one child.

$T_{\mathcal{A}}$ Infinite product tree generated by $\mathcal{A}$.

$(\mathcal{A}, \mathcal{G})$-**tree** Set-valued tree with root $\{I_d\}$ and internal nodes having either n positive children or at least one generator child.

Covered or uncovered node A node is covered if its set is a subset of an ancestor node; otherwise it is uncovered.

$\mathcal{L}(T)$ Leafage of T, i.e. all uncovered leaves.

1-**bounded** All matrix sets encoded by $\mathcal{L}(T)$ are bounded by 1 in a chosen matrix norm.

v-1-**bounded** All images of matrices encoded by $\mathcal{L}(T)$ applied to a fixed vector v are bounded by 1 in a chosen vector norm.

PL decomposition Prefix-leaf decomposition.

$T_{\min(\mathcal{A},\mathcal{G})}$ Minimal $(\mathcal{A}, \mathcal{G})$-tree, i.e. the smallest such tree containing a covered node.

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